Lab 6: Code and Branch Coverage Analysis

ATM Withdrawal Application

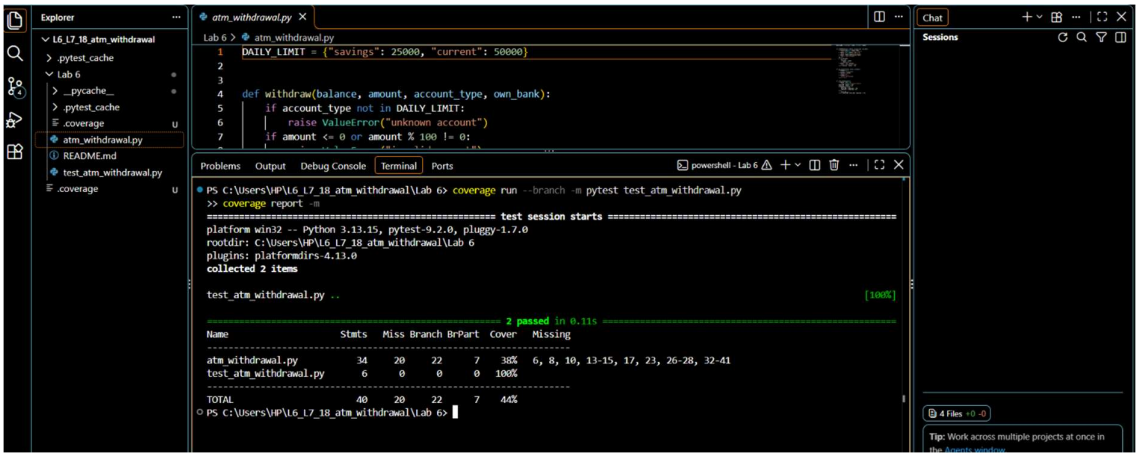
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1. Objective

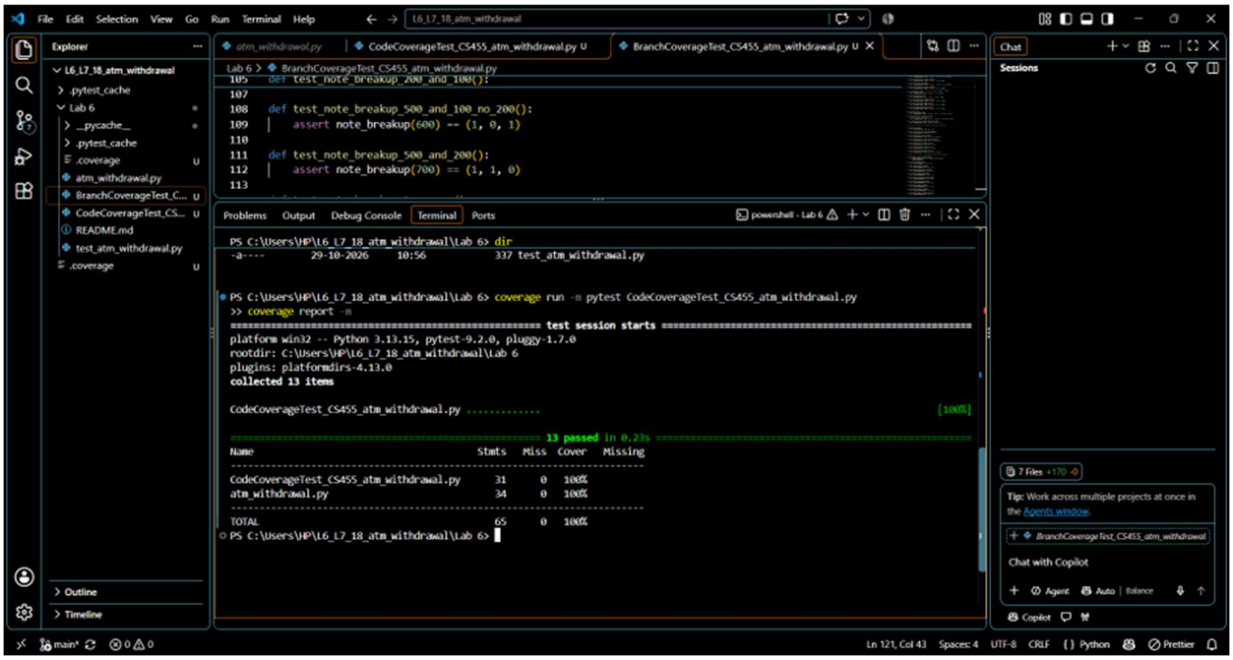
To measure line (code) and branch coverage of the ATM Withdrawal module (atm_withdrawal.py), write additional test cases for its conditions and edge cases, and reach at least 90% coverage for both.

The module has three functions: withdraw() (daily limits, fees, balance check), pin_check() (PIN attempts and card blocking) and note_breakup() (split of the amount into 500, 200 and 100 notes).



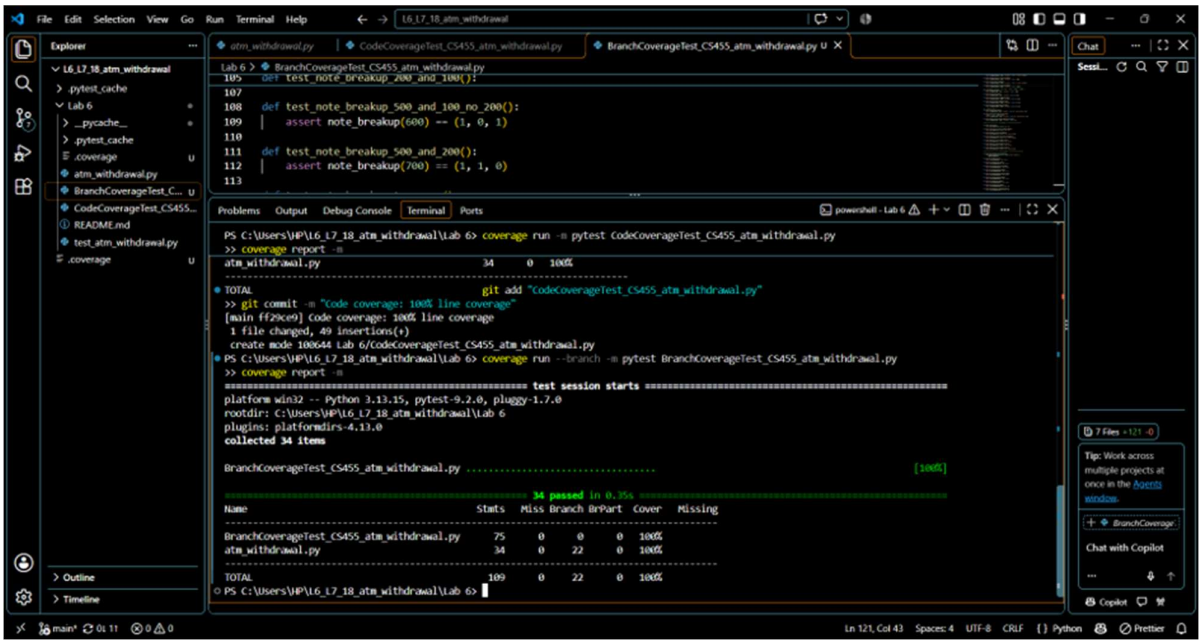
2. Code Coverage Results

Test file: CodeCoverageTest_CS455_atm_withdrawal.py



3. Branch Coverage Results

Test file: BranchCoverageTest_CS455_atm_withdrawal.py



Summary: 34 tests passed. atm_withdrawal.py has 34 statements and 22 branches, with 0 missed lines and 0 partial branches, giving 100% branch coverage.

4. Coverage Summary

Test file	Type	Coverage
test_atm_withdrawal.py (starter, 2 tests)	Starting point	38%
CodeCoverageTest_CS455_atm_withdrawal.py	Line coverage	100%
BranchCoverageTest_CS455_atm_withdrawal.py	Branch coverage	100%

5. What the Tests Cover

- withdraw(): unknown account, invalid amounts (zero, negative, not a multiple of 100), daily limits for savings and current accounts (exactly at and above the limit), own-bank and other-bank fees (20, and 30 above 10000), enough and insufficient balance.
- pin_check(): correct PIN, wrong PIN with retry, card blocked when attempts reach 3, and wrong PIN on the third try.
- note_breakup(): invalid amount and breakups using different mixes of 500, 200 and 100 notes.

6. Commands Used

```
pip install coverage pytest
coverage run -m pytest CodeCoverageTest_CS455_atm_withdrawal.py
coverage report -m
coverage run --branch -m pytest BranchCoverageTest_CS455_atm_withdrawal.py
coverage report -m
```

7. Conclusion

Adding targeted tests raised coverage of the ATM Withdrawal module from 38% to 100% for both line and branch coverage, exceeding the 90% requirement. Testing boundary values (such as exactly 25000 on a savings account and exactly 10000 for the fee threshold) made sure every decision branch was executed in both its True and False outcomes.

GitHub link: https://github.com/shamsanthosh123/L6_L7_18_atm_withdrawal